

Statistics Hybrid Practice Set – S9 (L5)

Topics

- Population
- Sample
- Mean
- Median
- Mode
- Range
- Population Variance
- Population Standard Deviation
- Quartiles (Q1, Q2, Q3)
- IQR
- Outlier Detection

Instructions

- Solve on **paper first**.
 - Then solve the **same questions in Excel**.
 - Write the Excel formula you used.
 - No calculator.
-

Q1. Identify Population and Sample

A bank has **18,500 customers**. It selects **600 customers** for a satisfaction survey.

Paper Task

- Population = ?
- Sample = ?

Excel Task

Total Customers	Surveyed Customers
18,500	600

Q2. Calculate the Mean

Value
25
30
35
40
45

Paper Task

- Calculate the Mean.

Excel Task

- Calculate the Mean.
 - Write the Excel formula.
-

Q3. Find the Median

Value
48
24
36
30
42

Paper Task

- Find the Median.

Excel Task

- Calculate the Median.
 - Write the Excel formula.
-

Q4. Find the Mode

Value
11
13
13
15
17
19

Paper Task

- Find the Mode.

Excel Task

- Calculate the Mode.
 - Write the Excel formula.
-

Q5. Calculate All Measures

Value
12
18
24
30
36

Paper Task

Calculate:

- Mean
- Range
- Population Variance
- Population Standard Deviation

Excel Task

Write all four formulas.

Q6. Quartiles & IQR

Value
6
8
10
12
14
16
18
20
22

Paper Task

Find:

- Q1
- Q2
- Q3
- IQR

Excel Task

Write all formulas.

Q7. Outlier Detection

Value
18
20
22
24
26
28
30
32
80

Paper Task

Find:

1. Q1
2. Q3
3. IQR
4. Lower Fence
5. Upper Fence
6. Is **80** an outlier?

Excel Task

Write formulas for:

- Q1
- Q3
- IQR

Q8. Interpretation

Two stores recorded daily sales.

Store A

Sales
98
100
101
99
102

Store B

Sales
50
75
100
125
150

Paper Task

Answer the following:

1. Which store has the **larger Range**?
2. Which store has the **higher Variance**?
3. Which store has the **higher Standard Deviation**?
4. Which store is **more consistent**?
5. Which measure would you use to compare consistency:

- Range
- Variance
- Standard Deviation

Explain your answer in one sentence.

Excel Task

Calculate:

- Range
- Population Variance
- Population Standard Deviation

for both stores.

Q9. Real Data Analyst Scenario

A retail company wants to identify **unusual sales values** before building a dashboard because extreme values may distort charts and averages.

Which statistical method is the **best** choice?

- A. Mean
- B. Standard Deviation
- C. IQR Method
- D. Mode

Explain your answer in one sentence.

Q10. Excel Formula Challenge

Write the Excel formulas for:

1. Mean
2. Median
3. Mode
4. Range
5. Population Variance
6. Population Standard Deviation
7. Q1
8. Q2
9. Q3
10. IQR

Use the range **B2:B10**.